

The coherence circle-packing lemma: an elementary uniform bound

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-10 · this version issued 2026-06-10 · v1.0

1. Purpose and scope: The gap this closes

The C_full off-diagonal closure (res5_033/035) rested on the antipodal lemma $T' \leq N/2$ with the packing $N \leq n_{\text{pack}} \approx 41$, giving $K_{\text{floor}} \leq 20$ and a THIN margin $\times 1.026$. That bound ignores a stronger constraint already in the admissible-class definition: *coherence resolution*. This note proves an elementary uniform bound from coherence that is tighter by a factor ~ 2 and makes the closure comfortable.

2. The coherence circle-packing lemma

Setup. The admissible class is coherence-resolved: any two points of Q are geodesically separated by $\geq \theta_{\min}$, the cap-packing bound being $n_{\text{pack}} = 16/\theta_{\min}^2$ (caps of radius $\theta_{\min}/2$). A sum-circle $C_t = \{x : x \cdot t = |t|^2/2\} \cap S^2$ ($t \neq 0$) is a Euclidean circle of radius $\rho = \sqrt{Q_0^2 - |t|^2/4} \leq Q_0$.

Lemma. For every admissible Q and every $t \neq 0$,

$$\boxed{\#(Q \cap C_t) \leq \left\lfloor \frac{2\pi}{\theta_{\min}} \right\rfloor} \quad \Rightarrow \quad T'(Q) = \max_{t \neq 0} \#(Q \cap C_t) \leq \left\lfloor \frac{2\pi}{\theta_{\min}} \right\rfloor.$$

Proof. Points of Q on C_t are pairwise geodesically $\geq \theta_{\min}$, hence pairwise Euclidean $\geq d := 2Q_0 \sin(\theta_{\min}/2)$. If k points lie on a circle of radius ρ with pairwise Euclidean distance $\geq d$, then the smallest adjacent gap satisfies $2\rho \sin(\pi/k) \geq d$, i.e. $k \leq \pi / \arcsin(d/(2\rho))$. Since $\rho \leq Q_0$,

$$\frac{d}{2\rho} \geq \frac{d}{2Q_0} = \sin \frac{\theta_{\min}}{2} \quad \Rightarrow \quad \arcsin \frac{d}{2\rho} \geq \frac{\theta_{\min}}{2} \quad \Rightarrow \quad k \leq \frac{2\pi}{\theta_{\min}}. \quad \blacksquare$$

The bound is geometric and uniform: it uses only the separation $\theta_{\min}$ and $\rho \leq Q_0$, with NO lattice / additive / number-theoretic input. It applies identically to lattice and non-lattice competitors.

3. The uniform numerical bound and adversarial check

Across the certified window $\theta_{\min} \in [0.596, 0.637]$, so $2\pi/\theta_{\min} \leq 10.54$ and

$$T'(Q) \leq 10 \quad \text{for every admissible competitor, uniformly on } \mathcal{R}.$$

Adversarial search (res5_036): a maximal admissible single-latitude-ring-pair reaches $T' = 8 < 10$ (near-great rings that would give higher occupancy violate coherence — the antipodal mirror collides near the equator); a random coherence-resolved config has $T' = 2$. The bound 10 is never exceeded.

4. Comfortable off-diagonal; the selection boundary binds

With R-025 ($K_{\text{floor}} \leq T'$):

$$K_{\text{floor}} \leq T' \leq 10 \Rightarrow R_{\text{lead}} = 23.2(1 + K_{\text{floor}})I \leq 23.2 \cdot 11 \cdot I.$$

At the operating intensity $I = 2 \times 10^{-3}$, $R_{\text{lead}} \leq 0.510$ (margin $\times 1.96$); at the selection cap $I_c^{\text{sel}} = 2.41 \times 10^{-3}$, $R_{\text{lead}} \leq 0.615 < 1$. The off-diagonal cap is

$$I_{\text{off}}^{\text{coh}} = \frac{1}{23.2 \cdot 11} = 3.92 \times 10^{-3} > I_c^{\text{sel}} = 2.41 \times 10^{-3},$$

so the SELECTION boundary binds (exactly as for the lattice class), and the C_full operating region is the FULL Step-1 selection region $\mathcal{R}$ (20% headroom at the operating point), NOT the thin off-diagonal-bound region (2.6%).

Consequence. The C_full off-diagonal margin is COMFORTABLE ($\times 1.96$), and the thin $\times 1.026$ of res5_033/035 is superseded as an artefact of the loose $T' \leq N/2$. The additive-energy extremal conjecture EXT (target $K_{\text{floor}} \sim 3$, margin $\times 5.4$) is now UNNECESSARY for comfort; it remains T2, an optional further tightening.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “Is $\theta_{\min}$ really the pairwise separation for all of Q ?” Yes — the admissible class is coherence-resolved with caps of radius $\theta_{\min}/2$ (centres $\geq \theta_{\min}$ apart); $n_{\text{pack}} = 16/\theta_{\min}^2$ is the matching cap count.
- (β) “Could a sum-circle have $\rho > Q_0$ and hold more points?” No — $\rho = \sqrt{Q_0^2 - |t|^2/4} \leq Q_0$ always; the great-circle limit ($\rho = Q_0$, $t \rightarrow 0$) is the worst case and gives the bound $2\pi/\theta_{\min}$.
- (γ) “Does the bound hold across the whole window?” Yes — $2\pi/\theta_{\min} \leq 10.54$ over the certified (I, μ^2) range (res5_036 scan), so $T' \leq 10$ uniformly; $\theta_{\min}$ grows with μ^2, I , tightening the bound.

Quantitative sanity check (mandatory): $\theta_{\min} = 0.627$, $2\pi/\theta_{\min} = 10.02$, $\lfloor \cdot \rfloor = 10$; window sup 10.54 $\Rightarrow T' \leq 10$; adversarial max 8; random 2; $R_{\text{lead}} \leq 0.510$ ($\times 1.96$); $I_{\text{off}}^{\text{coh}} = 3.92 \times 10^{-3} > I_c^{\text{sel}}$. All from res5_036 (5/5).

6. Result footer

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Result ID:          B1/B2 coherence circle-packing lemma --
                    comfortable C_full off-diagonal
Precise statement:   Coherence resolution (pairwise geodesic >=
                    theta_min) bounds every
                    sum-circle occupancy:  T'(Q) <=
                                         floor(2pi/theta_min) = 10
                                         uniformly across
                    the certified window (elementary circle-packing
                    proof; rho<=Q0). With
                    R-025 (K_floor<=T'):  R_lead <= 23.2*11*I =
                                         0.510 at I_op (x1.96),
                                         0.615 at
                    the selection cap -- COMFORTABLE, for lattice
                    AND non-lattice. I_off^coh=
                    1/(23.2*11)=3.92e-3 > I_c^sel=2.41e-3 =>
                    selection binds => C_full region
                    = full Step-1 region (20% headroom).
                    Adversarial max admissible T'=8;
                    random T'=2. Supersedes the thin x1.026 (loose
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Scope: $T' \leq N/2$ artefact).
 Off-diagonal sector for C_{full} , comfortable.
 EXT now optional (T2).
 Uniform across the certified window R .
 Elementary (no decoupling / NT).
 Dependencies: $R-025$ ($K_{\text{floor}} \leq T'$, T7); coherence-resolution
 admissibility ($n_{\text{pack}} =$
 $16/\theta_{\text{min}}^2$); Step 1 window ($I_{\text{c}}^{\text{sel}}$); T-018
 $(R_{\text{lead}}/K * \text{identity})$.
 Evidence grade: PROVED (elementary lemma) + EXECUTED (res5_036
 5/5).
 Reproduction command: python
 codes/vacuum/res5_036_coherence_offdiag_bound.py
 Expected output: $T' \leq \text{floor}(2\pi/\theta_{\text{min}}) = 10$; adversarial max
 8; random 2; $R_{\text{lead}} \leq 0.510$
 $(x1.96)$; selection binds ($I_{\text{off}}^{\text{coh}} > I_{\text{c}}^{\text{sel}}$);
 5/5 PASS.
 Falsifier: An admissible (coherence-resolved) competitor
 with a sum-circle of >10
 points; or $\theta_{\text{min}} < 2\pi/11 = 0.571$ somewhere
 in the window (would
 loosen the bound past 11).
 Tier before / after: Off-diagonal C_{full} : thin x1.026 (loose
 $T' \leq N/2$) -> COMFORTABLE x1.96
 (coherence $T' \leq 10$), selection-binding. EXT T2
 (now optional, not needed).
 No-overclaim statement: Bounds T' (hence K_{floor} via R-025) by
 coherence; comfortable but still
 scope-qualified (within R , given $A1$, C_{full}
 competitor class). Not a
 global/all-intensity claim. EXT (tight
 $K_{\text{floor}} \sim 3$) remains a separate T2.
 Next required action: Re-enact C_{full} as COMFORTABLE (supersede the
 thin-0 scope); G2 external
 referee (validation). EXT optional.